

8.3.9 Engine Compressor Rinsing

- CAUTION**
- SALT-LADEN HUMIDITY AND CHEMICALS WILL CORRODE COMPRESSOR BLADES AND VANES.
 - SALT-LADEN AIR MAY BE ENCOUNTERED UP TO 280 KM INLAND UNDER CERTAIN WEATHER CONDITIONS. IF ANY DOUBT EXISTS ABOUT THE CONDITIONS IN WHICH THE ENGINES WERE OPERATED, PERFORM A DAILY COMPRESSOR WATER RINSE.

Engines subjected to salt water contamination or other chemically laden atmosphere (including pesticides and herbicides) shall undergo water rinsing after shutdown following the last flight of the day.

The rinse operation (see BO 105 Maintenance Manual) should be performed as soon as practical after flight, but not before the engines have cooled to near ambient temperature.

8.4 COLD WEATHER OPERATION - GENERAL PRACTICES

8.4.1 Fuels

When operating in ambient temperatures below +4 °C (+40 °F) the fuel must contain an anti-icing additive meeting the requirements of MIL-I-27686.

If available use JP-4 or commercial Jet B fuel. As an alternate use an AVGAS/Jet fuel mixture following limitations described in Section 2, Fuel limitations.

8.4.2 Fuel/Additive Blending

- CAUTION** IF IT CANNOT BE DETERMINED THAT THE FUEL IN THE HELICOPTER CONTAINS AN ANTI-ICING ADDITIVE OR IF THE CONCENTRATION IS NOT KNOWN, THE HELICOPTER MUST BE DEFUELED AND REFUELED WITH FUEL CONTAINING THE PROPER ADDITIVE.

The required additive concentrations are in Section 2, Fuel limitations.

Blending the additive with the fuel is to be carried out using equipment normally supplied by the additive manufacturer. Follow manufacturer's recommended procedure when operating the blending equipment.

The flow of the additive must commence simultaneously with the flow of fuel. Refueling flow rates are to be between 95 and 227 liters per minute (25 and 60 US gal/min).

A differential refractometer may be used to accurately measure the concentration of additive in the fuel.

8.4.3 Mixing Alternate Fuel

CAUTION JP-4 OR COMMERCIAL JET B FUELS SHALL NOT BE MIXED WITH AVGAS.

As an alternate cold weather fuel when JP-4 or Commercial Jet B fuels cannot be obtained, use the following mixture:

- One part by volume AVGAS grade 80/87 or AVGAS 100L and,
- Two parts by volume Aviation Kerosene; JP-5; Commercial Jet A; or Commercial Jet A-1.

Refer to Fuel limitations in Section 2.

NOTE There is no engine operational time limit when using the AVGAS/Jet fuel mixture as long as 80/87 grade AVGAS is used and the 1:2 ratio is followed.

When operating on AVGAS/Jet fuel mixture, the anti-icing additive is required only for the jet fuel portion of the mixture.

CAUTION WHEN MIXING FUELS AVOID HIGH FLOW RATES. DO NOT EXCEED A FLOW RATE OF 189 LITERS PER MINUTE (50 US GAL/MIN). ALSO MAKE SURE THE FUEL NOZZLE AND FUEL TANK ARE ELECTRICALLY GROUNDED TO THE HELICOPTER.

Observe the following guidelines when making the alternate fuel mixture:

- Make the fuel mix in the helicopter tank. The AVGAS can be added before or after the jet fuel, the only restriction is that the final mix (including fuel in the tank before starting the mix) be two parts by volume jet fuel to one part by volume AVGAS.
- When the AVGAS/Jet fuel mixture is used in the engine the lead from the gasoline accumulates on the turbine and exhaust collector outlet ducts. The normal appearance after operation on lead fuel is a pale yellow powder deposit on the exhaust surfaces.
- If low temperature starting problems are encountered and the AVGAS/Jet fuel mixture is made in the fuel tank it may be necessary to bleed the unmixed fuel from the fuel lines in order to obtain a light-off. Refer to Allison Operation and Maintenance Manual.